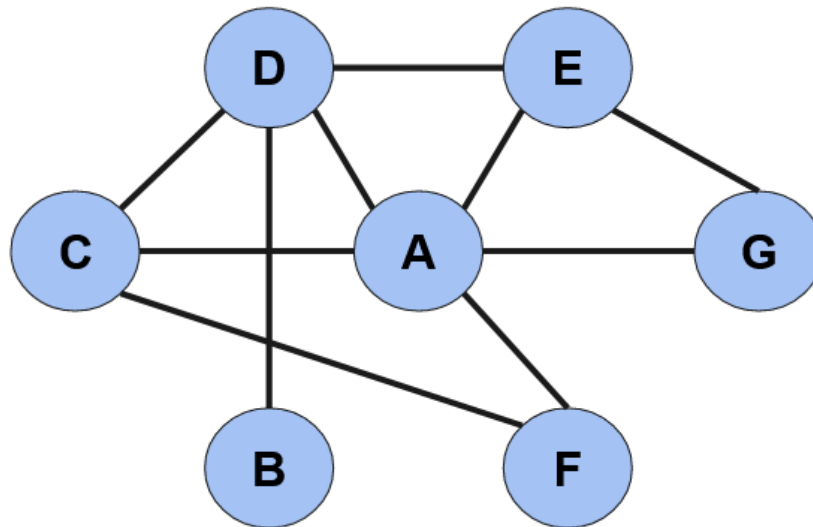


Practice Problem Set 4

1. Ego Network Composition (10pts)

1.



- a. Above is a picture of A's ego network. What is the clustering coefficient of A's ego network? (2pts)

- b. What is the ego adjacency matrix which reflects the connections between A's friends? (2pts)

- c. If A is a woman, and C, D, and B are also women, E and G are men, and F is nonbinary, what is the EI homophily of A in terms of gender identities? (2pts)

- d. We are interested in the diversity of A's network in terms of gender identities. What is A's Blau's heterogeneity index score? (2pts)
- e. What is the meaningful difference between EI homophily and Blau's heterogeneity index? (2pts)

2. *Let us imagine that Liz and Myles go to a school that is 35% green, 25% blue, 20% orange, and 20% purple. We are interested in category homophily and diversity. (12pts)*

- a. If Liz and Myles made 20 friends at random, what would be our best guess as to the category composition of their 20 friends? (2 pts)
- b. Liz is in the green group. What would Liz's Blau's heterogeneity index be if their friends were as described in part A? (2pts)
- c. What would Liz's EI homophily index be if their friends were as described in part A? (Liz is in the green group) (2pts)
- d. Myles is in the purple group. What would Myle's Blau heterogeneity index be if their friends were as described in part A? (2pts)

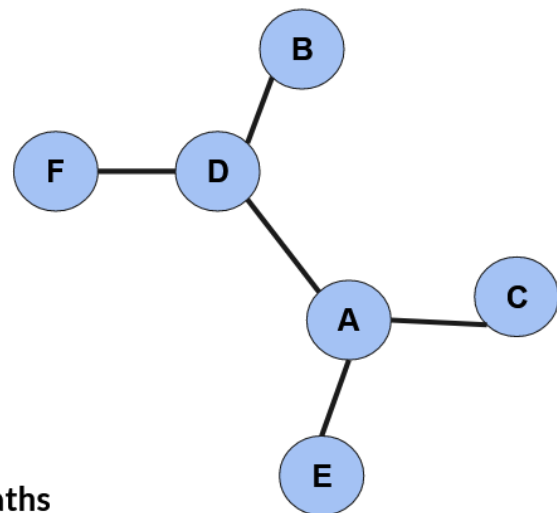
e. What is Myle's EI homophily index be if their friends were as described in part A (Myles is in the purple group) (2pts)

f. Does Liz have more or fewer in-group connections compared to Myles? Why? (2pts)

3. Fill in the distance matrix. What is the closeness centrality score for each of the nodes? (5pts)

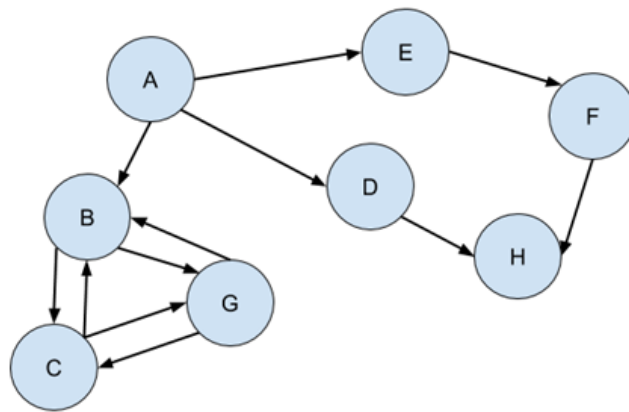
	<i>j</i>					
	A	B	C	D	E	F
<i>i</i> A	0					
B		0				
C			0			
D				0		
E					0	
F						0

Remember a distance matrix is shortest paths

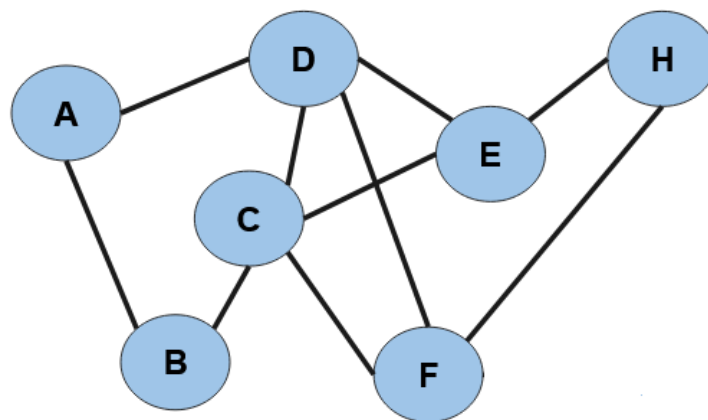


4. Answer the following questions about centrality.

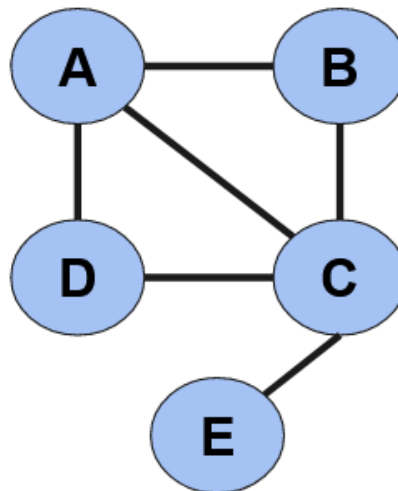
a) Which node has the highest out-degree centrality? (1pt)



b) Which node has the highest degree centrality? (1pt)



c) Which node has the highest betweenness centrality? (1pt)



5. Answer the following questions.

- a) Draw a graph with a structural hole. What is important about structural holes? (2pts)

- b) If node A is female and has 6 female friends, 2 male friends, and 2 non-binary friends, what is node A's relative rate homophily? How is this different from the expected rate? (2pts)

- c) Think about your personal life. What is the definition of a social foci and what social foci has played an interest on what friendship connections you have formed? (2pts)

- d) If you are interested in status, what centrality measure would you chose? What makes this measure useful for measuring the concept of status? (2pts)

- e) What is the difference between choice and induced homophily? (2pts)

- f) Draw a graph with one node with high betweenness centrality, and every other node with no betweenness centrality. What is the name for this type of graph? (2pts)